

A New Approach of Formation Control for Multi-Agent Systems With Environmental Changes

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Abstract—In this paper, a two-stage reconfiguration strategy is proposed for a group of agents to find their specially designated formation, which can be seen as a transition from the current states to the predefined formations. The specially designated formation occurs during the inter-task idle time in order to minimize the reconfiguration time. The basic specially designated formation is designed for dotted agents based on the optimal mapping relationships and current probabilities of the predefined formations. Then, the reconfiguration schemes are developed with the agents modeled as circles or ellipses to account for their physical shapes, and the problem of agent overlapping during the formation is addressed. Examples and simulation results are presented to analyze and demonstrate the effectiveness and potential of the new design techniques.

Index Terms—Multi-agent systems, reconfiguration strategy, random mapping algorithm, overlapping problem.

I. INTRODUCTION

THE problem of moving in formation for a group of agents has significant practical research value, especially in engineering field. Moving in group can reduce the system cost [1]–[3], improve the robustness [4], [5] and efficiency [6], [7] of the system while providing redundancy, reconfiguration ability and structure flexibility for the system [8]–[12]. This technique has been employed in controlling multiple underwater autonomous vehicles [13], [14], artificial unmanned vehicles [11], [15], mobile robots [16]–[18] and many other fields [19]–[23]. Reconfiguration ability is an essential feature of multi-agent system because it links to the problem of how many types of formations can be formed and transformed. In order to build the reconfiguration ability of multi-agent systems, reconfiguration strategies should be employed to drive the systems to construct different shapes of formation.

There is a lot of research on reconfiguration strategy for multiple agents [24]–[29]. In [24], a distributed cascade

robust feedback control scheme was built to drive a group of unmanned air vehicles which were vertical takeoff and landing to achieve the specific formation and reconfigured formation. The whole system was based on dynamic undirected communication network. The method based on coalition game theory and flocking-based formation maintenance mechanism was developed to solve the reconfiguration problem among multiple robots when they encounter obstacles in unknown environment in [30]. The problem of unsymmetrical formation reconfiguration and docking of multiple spacecrafts was studied in [25]. The compound optimal control was employed to deal with this problem with the total control effort and docking time, which were linked to the total fuel usage and electronic resources. In [26], a symplectic penalty iteration algorithm was proposed to achieve the optimal control for a group of spacecrafts, who had loose construction, with minimum energy consumption. In the algorithm, penalty functions were introduced to make sure the collision avoidance among spacecrafts. A continuous/impulsive linear quadratic regulator is constructed in [27]. It was used to design an optimal control scheme, which combined continuous Lorentz force actuation and impulsive thrusting, for spacecraft formation reconfiguration. Two-stage reconfiguration strategy was studied in [28], [29] for multiple aircrafts. The specially designated formation in [28] was obtained based on the reconfiguration energy consumption. The specially designated formation in [29], on the other hand, had the feature that the expected value of the reconfiguration time based on acceleration controller was minimized.

In [28], [29], the mapping relationships between the group of agents and the predefined formations have been regarded as fixed, which may result more consumption on group moving. Also, in the practical scenarios, agents always have certain shapes. The existing reconfiguration-related literature abstracts the agent as a dot agent, ignoring the overlapping problems caused by the characterization of circular agents and elliptical agents. In this paper, a two-stage reconfiguration strategy with a random mapping algorithm is proposed to find the optimal specially designated formation for a group of agents. The specially designated formation is a temporarily formation for the agents during the idle time, which is the time interval between the completion of the previous formation task and the invocation of the current reconfiguration task. In this reconfiguration scheme, the specially designated formation for dot agents is first designed based on the probability of the occurrence of the predefined formations and the optimal map-

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pings with the predefined formation set. The random mapping algorithm is employed to find the mapping relationships. The proposed strategy is updated for circular agents and elliptical agents due to the overlapping problem caused by the agent characterization. It is shown that the specially designated formation can minimize the expected reconfiguration time. After the next task command receiving, the group can move to the required formation by the specific formation algorithm.

There are real-world application areas the proposed two-stage reconfiguration strategy is needed, particularly when the involved vehicles are of considerable physical size. For example fire extinguishing UAVs are usually big for detecting and extinguishing bushfires. The predefined formations and the proposed strategy in this application can be obtained based on priori experience and the situation awareness of the fires. Other applications are in underwater, land and space exploration mission with multiple coordinating tasks for a group of physically large agents.

Our contributions in this paper are as follows. First, a two-stage reconfiguration strategy based on a group of dotted agents is proposed during the idle time. The mapping relationship between agents' current positions and each predefined formation is adaptive based on the operating distance. The specially designated formation of the dotted agents is constructed as the basic temporary formation for the following two strategies. Second, to meet the practical requirements, the reconfiguration scheme, which are used to find the optimal specially designated formation without overlapping problem during idle time, is developed due to the circular shapes of the agents. The overlapping agent set of each circular agent based on its characterization should be built to support the two-stage reconfiguration scheme. Third, the proposed reconfiguration strategy is updated for elliptical agents in view of their geometric features. It is important to coordinate the position and deviating direction of each elliptical agent to avoid overlapping problem in specially designated formation.

The rest of the paper is structured as follows. In Section II, the two-stage reconfiguration strategy for dot agents with random mapping algorithm is built. The two-stage reconfiguration strategy for circular and elliptical agents are introduced in Section III and Section IV. Simulation results are given in Section V. The paper is concluded in Section VI.

Notation: Throughout this paper, $\mathbf{R}^m$ is an m th dimensional space of real numbers. The symbol $|\cdot|$ represents the length of a vector. For any two points a and b , (a, b) represents the vector between a and b . Matrices are assumed to be compatible for algebraic operations. If the dimensions of matrices are not explicitly stated, they are assumed to be compatible for algebraic operations.

II. TWO-STAGE RECONFIGURATION STRATEGY FOR DOT AGENTS

In this section, a two-stage reconfiguration is developed with a random mapping algorithm for dot-shaped agents. A specially designated formation should be constructed during idle time T to simplify the movement process in next process. The position of each agent in this specially designated

formation can be obtained based on the probability of each predefined formation being the next task and the mapping relations of the current positions for all agents and each predefined formation. Idle time T is a period time between two movement process. In our two-stage reconfiguration strategy, it is assumed that idle time T is fixed and long enough to accomplish the calculation and movement to the specially designated formation. In this paper, n agents are assumed to form the multi-agent system E ,

$$E = \{E^1 \ E^2 \ \dots \ E^n\}. \quad (1)$$

The predefined formation set with ϱ predefined formation is given as

$$F = \{F^1 \ F^2 \ \dots \ F^\varrho\}, \quad (2)$$

while $F^s \in F$ represents the s th predefined formation which is given as

$$F^s = \{f_1^s \ f_2^s \ \dots \ f_n^s\} \in \mathbb{R}^{2n}. \quad (3)$$

In (2), f_i^s is the position of i th point in F^s , where $f_i^s \in \mathbb{R}^2$. The current positions for the multiple agents are given as

$$P = \{p_1 \ p_2 \ \dots \ p_n\} \in \mathbb{R}^{2n}, \quad (4)$$

where $p_j = (x_j^c, y_j^c)$, x_j^c and y_j^c are the x-axis coordinate and y-axis coordinate for agent E_j , respectively. The specially designated formation which is constructed during idle time T can be written as

$$P(a) = \{p_1(a) \ p_2(a) \ \dots \ p_n(a)\} \in \mathbb{R}^{2n}, \quad (5)$$

where $p_j(a) = (x_j^c(a), y_j^c(a))$, $x_j^c(a)$ and $y_j^c(a)$ are the x-axis coordinate and y-axis coordinate for agent E_j in specially designated formation, respectively. Instead of the fixed relation between agents and each predefined formation, the changing mappings are calculated to reduce the total moving distance. The random mapping algorithm is employed to obtain the optimal mappings for the current formation of the agents and each predefined formation. The random mapping algorithm is achieved by calculating and picking the minimum moving distance for the whole group of agents based on the random generated mapping in each calculation iteration. For the s th predefined formation F^s , to find the optimal mapping relation between F^s and the current position P , the random mapping algorithm is described as follows.

Initialization: In the first κ iterations, η mapping relations will be generated in each iteration. The concept of mapping relation used in this paper is the congruent relationship between each agent and the point in related predefined formation. In the k th iteration, the generated mapping relations are given as

$$M^s(k) = \{M_1^s(k) \ M_2^s(k) \ \dots \ M_\eta^s(k)\}, \quad 0 < k \leq \kappa, \quad (6)$$

$$M_i^s(k) = \{m_1^s(k) \ m_2^s(k) \ \dots \ m_n^s(k)\}, \quad 1 \leq i \leq \eta, \quad (7)$$

where, $M_i^s(k) \in M^s(k)$, is the i th generated mapping based on the s th predefined formation, while $m_i^s(k)$ is the homologous point in F^s of E_i in the k th iteration. The sum of the

distance between F^s and P based on each member in $M^s(k)$ is calculated by Euclidean distance, which is given as

$$L^s(k) = \{L_1^s(k) \ L_2^s(k) \ \dots \ L_\eta^s(k)\}, \quad (8)$$

$$L_i^s(k) = \sum_{j=1}^n |f_j^s(M_i^s(k)) - p_j|. \quad (9)$$

The optimal mapping $m_{op}^s(k)$ satisfies that

$$L_{op}^s(k) = \min\{L^s(k), L_{op}^s(k-1)\}, \quad (10)$$

where $L_{op}^s(k-1)$ represents the sum distance under the optimal mapping in the $(k-1)$ th iteration between F^s and P . For the first iteration, $L_{op}^s(k-1)$ can be calculated based on the initial mapping relation M_0^s which is given as a condition, while

$$\begin{aligned} M_0^s &= \{\bar{m}_1^s \ \bar{m}_2^s \ \dots \ \bar{m}_n^s\}, \quad 1 \leq s \leq \varrho, \\ M_0 &= \{M_0^1 \ M_0^2 \ \dots \ M_0^\varrho\}, \end{aligned}$$

where M_0 is the set of the initial mapping relations of the predefined formations.

In the first κ iterations, the optimal mapping in each iteration will be put into the optimal set $M_{op} \in \mathbb{R}^{n\eta}$,

$$M_{op} = \{m_{op}^s(1); m_{op}^s(2); \dots; m_{op}^s(\kappa)\}. \quad (11)$$

Then: From the $(\kappa+1)$ th iteration, the similarity of M_{op} should be detected first. In the $(\kappa+1)$ th iteration, if no coincident element occurs in M_{op} , which means the optimal mappings in the first κ iterations are totally different. Then η new mapping relations will be generated. The optimal mapping $m_{op}^s(\kappa+1)$ in the $(\kappa+1)$ th iteration could be filtered based on (8), (9) and (10), while the optimal set M_{op} will be changed to

$$M_{op} = \{m_{op}^s(2); m_{op}^s(3); \dots; m_{op}^s(\kappa+1)\}. \quad (12)$$

If there are overlapping elements in each row in M_{op} , the algorithm will act as follows. Assumed that r elements are the same in each row, if $r \ll n$, r can be regarded as 0. Then, the algorithm will run as if there are no duplicate elements. If $r < n$, then these overlapping elements can be seen as the fixed elements in optimal mapping $m_{op}^s(\kappa+1)$. Then, $(n-r)$ mappings with $(n-r)$ elements in each mapping should be generated, where $m_i^s(\kappa+1) \in \mathbb{R}^{(n-r)(n-r)}$, $0 < i \leq n-r$. Combined these $(n-r)$ elements and r fixed elements, based on (8), (9) and (10), optimal mapping $m_{op}^s(\kappa+1)$ can be obtained. Then the optimal set should be updated as (12). If all mappings in M_{op} are the same, the final optimal mapping m_{op}^s is written as $m_{op}^s(\kappa)$.

The random mapping algorithm will iterate until the final optimal mapping is found.

To acquire the position for each agent in the specially designated formation, the probability of occurrence of each predefined formation in the next mission should be given first. The probability of the s th predefined formation F^s is given as q^s , which represents the probability of predefined formation s occurs in the next mission. It can be seen that $q^s \in [0, 1]$, while $q^1 + q^2 + \dots + q^\varrho = 1$. Lemma 1 in [28] is used to find the specially designated formation for the group of dotted agents.

Lemma 1 ([28]): Consider a known finite set of points in $\mathbb{R}^2$, denoted by $F = \{p_i\}_{i=1}^\varrho$, and a given set of weights $Q = \{q_i\}_{i=1}^\varrho$, where $\sum_{i=1}^\varrho q_i = 1$. Define the point $P = \sum_{i=1}^\varrho q_i p_i$ as the weighted center of gravity of F . Define also $S_\chi = \sum_{i=1}^\varrho q_i \|d_i - \chi\|^2$, where $\|\cdot\|$ denotes the Euclidean norm and χ is an arbitrary vector in $\mathbb{R}^2$. The minimum value of S_χ occurs when $\chi = P$. Moreover, $S_\chi = S_P + \|P - \chi\|^2$, where $S_\chi = \sum_{i=1}^\varrho q_i \|d_i - P\|^2$.

By using Lemma 1, the position for each agent in the optimal specially designated formation can be obtained by

$$p_j(a) = \sum_{i=1}^\varrho q^i f_j^i, \quad 1 \leq j \leq n, \quad (13)$$

where q^i is the probability of occurrence of the agent j in predefined formation F^i . The position corresponding to E_j based on m_{op}^i in the i th predefined formation. In this section, the agents are considered as dots, which means no collision will occur among the agents. Hence, the positions for all agents in the group based on (13) are the final positions in specially designated formation. The algorithm for the two-stage reconfiguration algorithm for a group dot agents is given as follow.

Algorithm 1 Two-Stage Reconfiguration for Dot Agents

Input:

The set of predefined formation, F ;
The set of the current positions of the centers of the dot-shape agents, P ;
The number of the predefined formation, ϱ ;
The number of the agent, n ;
The coefficient, κ ;
Initial mapping relation sets, M_0 ;
Probability of each predefined formation, $q = \{q^1 \ q^2 \ \dots \ q^\varrho\}$;

Output:

Find the specially designated formation during idle time T , $P(a)$;

```

1: for  $1 \leq s \leq \varrho$  do
2:   while do
3:     for  $k = 1 : 10000000$  do
4:       if  $k \leq \kappa$  then
5:         Generate  $\eta$  mapping relations,
6:          $M^s(k) = \{M_1^s(k) \ M_2^s(k) \ \dots \ M_\eta^s(k)\}$ ;
7:         for  $1 \leq i \leq \eta$  do
8:           Calculate the sum distance based on  $M_i^s$ ;
9:            $L_i^s(k) = \sum_{j=1}^n |f_j^s(M_i^s(k)) - p_j|$ ;
10:        end for
11:        $L_{op}^s(k) = \min\{L_1^s(k) \ L_i^s(k) \ \dots \ L_\eta^s(k), L_{op}^s(k-1)\}$ ;
12:       Find  $m_{op}^s(k)$  based on  $L_{op}^s(k)$ ;
13:        $M(op)(k, :) = m_{op}^s(k)$ ;
14:     else
15:       Find the number of overlapping elements in
16:        $M(op)$ ,  $r$ ;
17:       if  $r = 0$  or  $r \ll n$  then
18:         Repeat the steps such as  $k \leq \kappa$ ;
19:         Update  $M(op)$ ;
20:       else if  $r < n$  then
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20:       $r$  overlapping elements is seen as the fixed
      elements in  $m_{op}^s(k)$ ;
21:      Generate  $n-r$  mappings with  $n-r$  elements;
22:      Combined  $n-r$  generated elements and  $r$ 
      fixed elements to find  $L_{op}^s(k)$ ;
23:      Find  $m_{op}^s(k)$  based on  $L_{op}^s(k)$ ;
24:      Update  $M(op)$ ;
25:   else
26:       The final optimal mapping  $m_{op}^s = m_{op}^s(\kappa)$ ;
27:       break;
28:   end if
29: end if
30: end for
31: end while
32: end for
33: for  $1 \leq j \leq n$  do
34:    $p_j(a) = \sum_{i=1}^q q^i f_j^i$ ;
35: end for
36:  $P(a) = \{p_1(a) \ p_2(a) \ \dots \ p_n(a)\}$ 

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III. TWO-STAGE RECONFIGURATION STRATEGY FOR CIRCULAR AGENTS

In this section, the two-stage reconfiguration strategy for circular agents is developed to investigate the overlapping problem. In reality, agents always have their own shapes instead of dots, which will lead to collision and overlapping problems. To solve the possible overlapping issue, the two-stage reconfiguration strategy for circular agents is proposed to find the optimal specially designated formation $P(c) = \{p_1(c) \ p_2(c) \ \dots \ p_n(c)\}$ for the group.

Firstly, $P(a)$ in idle time T can be calculated by (13). The distance between each point in $P(a)$ should be calculated to determine whether the circular agents overlap. The distance set $D \in \mathbb{R}^{nn}$ can be calculated as

$$d_{jl} = |p_j - p_l|, \quad j \neq l, \quad d_{jl} = \infty, \quad j = l, \quad (14)$$

where d_{jl} is the distance between the positions p_j and p_l of agent j and agent l in $P(a)$, while p_j and p_l can be obtained by (13). To avoid overlapping among circular agents, reference distance set G is given as

$$G = \begin{bmatrix} g_{11} & g_{12} & \cdots & g_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ g_{n1} & g_{n2} & \cdots & g_{nn} \end{bmatrix},$$

$$g_{jl} = r_j + r_l + \varepsilon, \quad j \neq l,$$

$$g_{jl} = 0, \quad j = l, \quad (15)$$

where r_j and r_l are the radius of circular E_j and E_l , respectively. Coefficient ε is a constant parameter, while d_{jl} is the reference coincidence distance for agent j to agent l . The difference between G and D is given as

$$\tilde{D} = G - D, \quad \tilde{d}_{jl} = g_{jl} - d_{jl}. \quad (16)$$

Based on $\tilde{D}$, it can be determined whether there is overlap among circular agents.

Theorem 1: Based on $P(a)$ obtained in (13), for circular agent j , if all $\tilde{d}_{jl} \in \tilde{D} \leq 0$, the position for agent j in $P(c)$ can be obtained as

$$p_j(c) = p_j(a),$$

where the position of E_j in $P(c)$ is $p_j(c)$ and the position of E_j in $P(a)$ is $p_j(a)$.

Proof: For circular agent j , if all $\tilde{d}_{jl} \in \tilde{D} \leq 0$, $E_l \in E$, we have $g_{jl} \leq d_{jl}$, which means all actual distances between circular agent j and the other circular agents not less than the reference distances given in (15). Hence, agent j does not have overlapping problem. The position $p_j(c)$ of agent j in $P(c)$ does not need to be updated based on $p_j(a)$, which is given as

$$p_j(c) = p_j(a). \quad (17)$$

□

If there is $\tilde{d}_{jl} \in \tilde{D} > 0$, there is overlapping problem for E_j . Therefore, the position of E_j in $P(c)$ should be updated based on it in $P(a)$. Assumed that α agents coincide with E_j , the overlapping agent set of E_j is given as

$$E^j(P(c)) = \{E_1^j(P(c)) E_2^j(P(c)) \dots E_\alpha^j(P(c))\}. \quad (18)$$

To analyze the overlapping direction, the following decision method is proposed

$$\begin{cases} S_1 = \{(x_l^c(a) - x_j^c(a))\}, \\ S_2 = \{(y_l^c(a) - y_j^c(a))\}, \end{cases} \quad (19)$$

where $E_l \in E^j(P(c))$, and $x_l^c(a)$, $x_j^c(a)$, $y_l^c(a)$ and $y_j^c(a)$ are the x-axis and y-axis coordinates of E_j and E_l in $P(a)$, respectively. If there is $s_1, s_2 \in S_1$ satisfy

$$s_1 \times s_2 > 0, \quad (20)$$

or there is $s_3, s_4 \in S_2$ satisfy

$$s_3 \times s_4 > 0, \quad (21)$$

the overlapping areas of the E_j and all $E_l \in E^j(P(c))$ is on one side. The positions $p_j(c)$ of E_j and $p_l(c)$ of E_l satisfies above situation can be obtained by Theorem 2.

Theorem 2: Based on $P(a)$ obtained in (13), for circular agent j , if $p_j(a)$ satisfies (20) or (21) with $p_l(a)$ of $E_l \in E^j(P(c))$. The positions of E_j and E_l in $P(c)$ can be obtained as follows:

$$\begin{cases} p_j(c) = \begin{cases} p_j(a) - \mu(g_{jl} - d_{jl}) \cos \theta_{jl} \vec{u}, \\ s_1 \times s_2 > 0 \\ p_j(a) - \mu(g_{jl} - d_{jl}) \sin \theta_{jl} \vec{v}, \\ s_3 \times s_4 > 0, \end{cases} \\ p_l(c) = p_l(a), \end{cases}$$

where $p_j(c)$ is the position of E_j in $P(c)$ and $p_j(a)$ is the position of E_j in $P(a)$. Distances d_{jl} and g_{jl} are obtained in (14) and (15), and θ_{jl} is the angle between $(\vec{p}_j(a), \vec{p}_l(a))$ and axis x or y based on S_1 and S_2 . The parameter μ satisfies $0 < \mu < 0.5$. To calculate $p_j(c)$, E_l is the agent with the maximum distance on the moving direction of E_j , while the positions of E_j and E_l are given as $p_j(a) = (x_j^c(a), y_j^c(a))$

and $p_l(a) = (x_l^c(a), y_l^c(a))$, and parameters $\vec{u}$ and $\vec{v}$ are unit vectors.

Proof: if any $s_1, s_2 \in S_1$, $s_1 \times s_2 > 0$ or any $s_3, s_4 \in S_2$, $s_3 \times s_4 > 0$, then it can be seen that all agents that overlapped with E_j are on one side of agent j . Coincidence distances W_j of E_j and $E_l \in E^j(P(c))$ can be obtained as

$$W_j = \{w_1 \ w_2 \ \dots \ w_\alpha\}, \quad w_l = g_{jl} - d_{jl}, \quad 1 \leq l \leq \alpha, \quad (22)$$

where g_{jl} and d_{jl} can be obtained in (14) and (15).

If $s_1 \times s_2 > 0$, then W_j will be projected onto the x-axis as

$$H_j = \{h_1 \ h_2 \ \dots \ h_\alpha\}, \quad h_l = w_l \cos \theta_{jl}, \quad 1 \leq l \leq \alpha, \quad (23)$$

where θ_{jl} is the angle between $\overrightarrow{(p_j(a), p_l(a))}$ and x-axis,

$$\theta_{jl} = \arccos \frac{x_l^c(a) - x_j^c(a)}{|p_l(a) - p_j(a)|}, \quad E_l \in E^j(P(c)), \quad (24)$$

where $p_j(a)$ and $p_l(a)$ are the positions of E_j and E_l in $P(a)$, respectively. And $x_j^c(a)$ and $x_l^c(a)$ are the coordinates of these two agents in x-axis.

Based on (23) and (24), moving vector λ_j for agent j is written as

$$\lambda_j = |\lambda_j| \vec{u}, \quad (25)$$

where $|\lambda_j|$ represents the length of λ_j for E_j . It can be obtained by

$$\begin{aligned} |\lambda_j| &= \mu \max H_j \\ &= \pm \mu (g_{jl} - d_{jl}) \cos \theta_{jl}, \end{aligned} \quad (26)$$

where $0 < \mu < 0.5$ is a constant coefficient, and E_l is corresponded to $\max D_p^j$. Symbol $\pm$ is decided by θ_{jl} . The unit vector that $\overrightarrow{(p_j(a), p_l(a))}$ projects onto the x-axis is $\vec{u} = [1, 0]$ and $-\vec{u} = [-1, 0]$, which is decided by θ_{jl} . If all $x_l^c(a) - x_j^c(a) > 0$, all $E_l \in E^j(P(c))$ are on the right side of E_j . Thus, $-\pi/2 < \theta_{jl} < \pi/2$, and the moving direction of E_j is along $[-1, 0]$. The moving distance is given as

$$|\lambda_j| = \mu (g_{jl} - d_{jl}) \cos \theta_{jl}. \quad (27)$$

If all $x_l^c(a) - x_j^c(a) < 0$, the circular agents in $E^j(P(c))$ distribute in the left side of E_j , while $\pi/2 < \theta_{jl} < 3\pi/2$, and $\vec{u} = [1, 0]$. The moving distance for E_j is obtained by

$$|\lambda_j| = -\mu (g_{jl} - d_{jl}) \cos \theta_{jl}. \quad (28)$$

If $s_3 \times s_4 > 0$, the moving direction is on y-axis, the moving distance for E_j is written as

$$|\lambda_j| = \pm \mu (g_{jl} - d_{jl}) \sin \theta_{jl}, \quad (29)$$

where E_l corresponds to $\max D_p^j$, and $0 < \mu < 0.5$. The direction vector e_{jl} is written as $\vec{v} = [0, 1]$ and $-\vec{v} = [0, -1]$. When all $E_l \in E^j(P(c))$ satisfy $y_l^c(a) - y_j^c(a) > 0$, all E_l are above E_j . Angles θ_{jl} satisfy $0 < \theta_{jl} < \pi$, and the projected direction is $-\vec{v}$. The moving distance is obtained by

$$|\lambda_j| = \mu (g_{jl} - d_{jl}) \sin \theta_{jl}. \quad (30)$$

If all $y_l^c(a) - y_j^c(a) < 0$, all E_l are below agent j , and $-\pi < \theta_{jl} < 0$. Hence, projected direction is $\vec{v}$, and the moving distance is written as

$$|\lambda_j| = -\mu (g_{jl} - d_{jl}) \sin \theta_{jl}. \quad (31)$$

Based on above derivation, the position of E_j in $P(c)$ can be written as

$$p_j(c) = \begin{cases} p_j(a) - \mu (g_{jl} - d_{jl}) \cos \theta_{jl} \vec{u}, & s_1 \times s_2 > 0 \\ p_j(a) - \mu (g_{jl} - d_{jl}) \sin \theta_{jl} \vec{v}, & s_3 \times s_4 > 0, \end{cases} \quad (32)$$

where E_l is corresponding to the maximum moving distance in moving direction. Unit vectors $\vec{u}$ and $\vec{v}$ satisfy $\vec{u} = [1, 0]$ and $\vec{v} = [0, 1]$.

The position $p_l(c)$ of $E_l \in E^j(P(c))$ in $P(c)$ is same with $p_l(a)$, which is given as

$$p_l(c) = p_l(a). \quad (33)$$

□

Based on (19), for circular agent j , if there are multiple $p_l(a)$ of $E_l \in E^j(P(c))$, satisfy $s_1, s_2 \in S_1$, $s_1 \times s_2 < 0$ and $s_3, s_4 \in S_1$, $s_3 \times s_4 < 0$, E_j exists overlapping problem, and the overlapping areas surround E_j . The positions $p_j(c)$ of E_j and $p_l(c)$ of E_l around E_j can be obtained by Theorem 3.

Theorem 3: Based on $P(a)$ obtained in (13), for circular agent j , if $p_j(a)$ satisfies (20) or (21) with $p_l(a)$ of $E_l \in E^j(P(c))$. The positions of E_j and E_l in $P(c)$ can be obtained as follows:

$$\begin{cases} p_j(c) = p_j(a), \\ p_l(c) = p_l(a) + \mu (g_{jl} - d_{jl}) \frac{p_l(a) - p_j(a)}{|p_l(a) - p_j(a)|}, \end{cases}$$

where $p_j(c)$ is the position of E_j in $P(c)$ and $p_j(a)$ is the position of E_j in $P(a)$. Distances d_{jl} and g_{jl} are obtained in (14) and (15). The parameter μ satisfies $0 < \mu < 0.5$. To calculate $p_j(c)$, E_l is the agent with the maximum distance on the moving direction of E_j , while the positions of E_j and E_l are given as $p_j(a) = (x_j^c(a), y_j^c(a))$ and $p_l(a) = (x_l^c(a), y_l^c(a))$.

Proof: For circular agent j , if there are multiple $p_l(a)$ of $E_l \in E^j(P(c))$, satisfy $s_1, s_2 \in S_1$, $s_1 \times s_2 < 0$ and $s_3, s_4 \in S_1$, $s_3 \times s_4 < 0$ in (19), the overlapped agents of E_j surround E_j . The position of E_j in $P(c)$ keeps the original position in $P(a)$,

$$p_j(c) = p_j(a). \quad (34)$$

For $E_l \in E^j(P(c))$, the moving vector λ_l can be obtained by

$$\lambda_l = |\lambda_l| e_{jl}, \quad (35)$$

where $|\lambda_l|$ is the length of moving distance of E_l , and e_{jl} is the unit vector of moving direction.

$$|\lambda_l| = \mu d_l = \mu (g_{jl} - d_{jl}), \quad (36)$$

$$e_{jl} = \frac{p_l(a) - p_j(a)}{|p_l(a) - p_j(a)|}, \quad (37)$$

where w_l is the maximum overlapping distance between E_l and E_j . The center positions of E_j and E_l are represented by p_j and p_l , respectively. Parameter μ satisfies $0 < \mu < 0.5$. Hence the positions of E_j and $E_l \in E^j(P(c))$ are given as

$$p_j(c) = p_j(a). \quad (38)$$

$$p_l(c) = p_l(a) + \mu (g_{jl} - d_{jl}) \frac{p_l(a) - p_j(a)}{|p_l(a) - p_j(a)|}. \quad (39)$$

□

It can be seen that the algorithm is convergent. The sum of the overlapping areas in time t of the whole group can be defined as $Z(t)$, while

$$Z(t) = \sum_{i=1}^n z_i(t),$$

where $z_i(t)$ represents the overlapping area on E_i in t . The set of agents that appears to coincide in t is defined as $E^O(t) = \{E_1^O(t), E_2^O(t), \dots\}$, $E^O(t) \in E$. Then $Z(t)$ can be expressed as

$$Z(t) = \sum_{i=1}^{|E^O(t)|} z_{E_i^O(t)}(t).$$

If $Z(t) = 0$, which means circular agents in t do no overlap with each other, the two-stage reconfiguration strategy will stop, and the formation in t will be treated as the specially designated formation for circular agents.

If $Z(t) > 0$, $Z(t+1)$ is expressed as

$$Z(t+1) = \sum_{i=1}^{|E^O(t+1)|} z_{E_i^O(t+1)}(t+1),$$

where $|E^O(t+1)| \leq |E^O(t)|/2$, and $E_i^O(t+1) \in E^O(t)$. Based on Theorem 1, if overlapping with other agents during the movement does not occur, thus $z_{E_i^O(t+1)} + z_{E_j^O(t+1)} \leq z_{E_i^O(t+1)} + z_{E_j^O(t+1)}$, $E_i^O(t+1) \in E^O(t)$, $E_j^O(t+1) \in E^O(t)$. It can be seen that E_i and E_j have an overlapping problem in time t . Inequality holds due to the parameter μ satisfying $0 < \mu < 0.5$. If the overlapping problem with other agents occurs while moving, we have $z_{E_i^O(t+1)} + z_{E_j^O(t+1)} + z_{E_l^O(t+1)} \leq z_{E_i^O(t+1)} + z_{E_j^O(t+1)}$, in which $E_l^O(t+1) \in E^O(t)$, $E_j^O(t+1) \in E^O(t)$, and E_l is newly added agent with overlapping relationship with E_i and E_j . Owing to the opposite directions of movement of E_i and E_j , E_l can only overlap with one. Assumed E_i overlap with E_l in $t+1$, the maximum overlapping distance $w_l(t+1)$ between them can be obtained from (22), and satisfies

$$w_l(t+1) \leq |\lambda_i(t)| + |\lambda_l(t)| - \tilde{d}_{il}(t),$$

where $|\lambda_i(t)|$ and $|\lambda_l(t)|$ are calculated based on Theorem 1, and $\tilde{d}_{il}(t)$ is obtained by (16). Based on this inequality, we have $z_{E_i^O(t+1)} + z_{E_j^O(t+1)} + z_{E_l^O(t+1)} \leq z_{E_i^O(t+1)} + z_{E_j^O(t+1)}$. Then, we have

$$Z(t+1) \leq \sum_{i=1}^{|E^O(t+1)|} z_{E_i^O(t+1)}(t) \leq Z(t).$$

Thus, the two-stage reconfiguration algorithm for circular agents is convergent.

IV. TWO-STAGE RECONFIGURATION STRATEGY FOR ELLIPTICAL AGENTS

In this section, the two-stage reconfiguration strategy for a group of elliptical agents is established. The agents in the group are modeled as ellipses because many practical agents have a long and narrow shape. The objective of the

two-stage reconfiguration strategy for elliptical agents is to update the positions in $P(a)$ obtained in Section II to find the specially designated formation $P(e)$ for elliptical agents. The specially designated formation is written as $P(e) = \{p_1(e) p_2(e) \dots p_n(e)\}$. The updated reconfiguration scheme is used to avoid overlapping problem among elliptical agents. The difficulty addressed in this section is the decision method of coincident of elliptical agents and the maximum overlap distance for two elliptical agents. First, the formula of elliptical agent j is given as

$$E_j : [x_j, y_j, 1] A_j [x_j, y_j, 1]^T = 0 \quad (40)$$

where

$$A_j = \begin{bmatrix} \frac{\cos^2 \phi_j}{a_j^2} + \frac{\sin^2 \phi_j}{b_j^2} & \frac{\sin 2\phi_j}{a_j^2} & \frac{2A_{j1} \cos \phi_j}{a_j^2} \\ -\frac{\sin 2\phi_j}{b_j^2} & \frac{\sin^2 \phi_j}{a_j^2} + \frac{\cos^2 \phi_j}{b_j^2} & \frac{2A_{j1} \sin \phi_j}{a_j^2} \\ -\frac{2A_{j2} \sin \phi_j}{b_j^2} & \frac{2A_{j2} \cos \phi_j}{b_j^2} & \frac{A_{j1}^2}{a_j^2} + \frac{A_{j2}^2}{b_j^2} - 1 \end{bmatrix},$$

where A_j is the parameter matrix based on heading angle ϕ_j with

$$\begin{aligned} A_{j1} &= x_j^c \cos \phi_j + y_j^c \sin \phi_j, \\ A_{j2} &= -x_j^c \sin \phi_j + y_j^c \cos \phi_j. \end{aligned}$$

In A_j , a_j represents the long axes of the E_j , and b_j is the short axes. The coordinate of center of E_j given in (4) is given as (x_j^c, y_j^c) . The set of points on E_j is described as follow,

$$\begin{aligned} \hat{P}_j &= \{\hat{p}_j = (x_j, y_j) | \\ x_j &= a_j \cos \vartheta \cos \phi_j - b_j \sin \vartheta \sin \phi_j + x_j^c, \\ y_j &= a_j \cos \vartheta \sin \phi_j + b_j \sin \vartheta \cos \phi_j + y_j^c \}, \end{aligned} \quad (41)$$

where $\vartheta \in [0, 2\pi]$. Based on the $P(a)$, the following equations for E_j and E_l should be solved to make the decision on whether there is an overlapped problem,

$$\begin{cases} [x_j(a), y_j(a), 1] A_j [x_j(a), y_j(a), 1]^T = 0, \\ [x_l(a), y_l(a), 1] A_l [x_l(a), y_l(a), 1]^T = 0. \end{cases} \quad (42)$$

Four overlapping cases will occur during solving (42). For elliptical agent j , if there is no real solution in (42) for any $E_l \in E$, $E_l \neq E_j$, and $\hat{p}_j \in \hat{P}_j$ satisfies

$$[\hat{p}_j, 1] A_l [\hat{p}_j, 1] > 0,$$

or $\hat{p}_l \in \hat{P}_l$ satisfies

$$[\hat{p}_l, 1] A_j [\hat{p}_l, 1] > 0,$$

then there is no intersection between E_l and E_j . Hence, the position of E_j in $P(e)$ can be obtained as

$$p_j(e) = p_j(a), \quad (43)$$

where $p_j(e)$ is the position of E_j in $P(e)$ and $p_j(a)$ is the position of E_j in $P(a)$.

If there is no real solutions in (40) for any $E_l \in E$, and $\hat{p}_j \in \hat{P}_j$ satisfies

$$[\hat{p}_j, 1] A_l [\hat{p}_j, 1] < 0,$$

or $\hat{p}_l \in \hat{P}_l$ satisfies

$$[\hat{p}_l, 1]A_j[\hat{p}_l, 1] < 0,$$

then E_j and E_l are inclusion relationship in $P(a)$.

If there are more than one real solution in (40) for $E_l \in E$, the overlapped problem is existed in $P(a)$. Assumed β agents coincide with E_j (contain inclusion relationship), the overlapping set for E_j is written as

$$E^j(P(e)) = \{E_1^j(P(e)) E_2^j(P(e)) \dots E_\beta^j(P(e))\}. \quad (44)$$

Based on this situation, the overlapping problems of E_j are twofold: i) Based on (18) in Section III, the surrounding situations of the overlapping direction can be obtained as follows: if any $s_1 \in S_1, s_2 \in S_1, s_1 \times s_2 > 0$ or any $s_3 \in S_2, s_4 \in S_2, s_3 \times s_4 > 0$, then the overlapping areas of the E_j and all $E_l \in E^j(P(e))$ are on one side; ii) If there exist $s_1 \in S_1, s_2 \in S_1, s_1 \times s_2 < 0$ and $s_3 \in S_2, s_4 \in S_2, s_3 \times s_4 < 0$, the intersections of E_j and $E_l \in E^j(P(e))$ are around E_j .

If the overlapping areas of E_j are distributed on one side of E_j , the positions for elliptical agent E_j in $P(e)$ can be updated based on the Theorem 4 as follow.

Theorem 4: Let the elliptical agents satisfied (40) be currently positioned in P . Based on $P(a)$ obtained in (13), the positions for elliptical agents E_j and $E_l \in E^j(P(e))$ in $P(e)$ can be obtained as follows:

$$\begin{cases} p_j(e) = \begin{cases} p_j(a) - (\mu\lambda_{jl} \cos\theta_{jl} + \varepsilon)\vec{u}, & s_1 \times s_2 > 0 \\ p_j(a) - (\mu\lambda_{jl} \sin\theta_{jl} + \varepsilon)\vec{v}, & s_3 \times s_4 > 0, \end{cases} \\ p_l(e) = p_l(a), \end{cases}$$

where E_l is the elliptical agent with the maximum distance on the moving direction of E_j , while $p_j(e)$ is the position of E_j in $P(e)$ and $p_j(a)$ is the position of E_j in $P(a)$. Vector λ_{jl} is the moving distance of E_j , and θ_{jl} is the projection angle of λ_{jl} to x-axis. The constant parameter μ is used to control the length of moving distance, and $0 < \mu < 0.5$. Coefficient ε is proposed to ensure the moving distance.

Proof: Under this condition, for elliptical agent E_j , all intersections of E_j are on one side of E_j . Moving vector λ_j of E_j can be obtained by

$$\lambda_j = (\mu|\lambda_j| + \varepsilon)e_{jl}, \quad (45)$$

where $|\lambda_j|$ is the moving distance of E_j , and e_{jl} is the moving direction. Parameter μ is the positive coefficient, and $0 < \mu < 0.5$, and ε is a positive coefficient to ensure the moving length of E_j . Moving distance $|\lambda_j|$ can be obtained based on the conditions of overlapping distribution. If $s_1 \times s_2 > 0, s_1 \in S_1, s_2 \in S_1$, the intersections are distributed to the left or right of E_j , and $|\lambda_j|$ can be written as

$$|\lambda_j| = \max\{|\lambda_{jl}| \cos\theta_{jl}\}, E_l \in E^j(P(a)), \quad (46)$$

where λ_{jl} is the moving distance of E_j based on E_l , while θ_{jl} is the angle between moving direction and x-axis.

There are three situations to calculate λ_{jl} and θ_{jl} . The first situation is that there are more than two real solutions of (40), which means two agents have more than two points of intersection. Second situation is the inscribed situation,

which is given as: there are one or two points of intersection $\hat{p}_1 = (x_1, y_1)$ and $\hat{p}_2 = (x_2, y_2)$, and there are

$$\begin{aligned} d_f(x, y) &= \sqrt{(x - x_l^c(a))^2 + (y - y_l^c(a))^2}, \\ d_f(\hat{p}_1) &> d_f(\hat{p}_j), \\ d_f(\hat{p}_2) &> d_f(\hat{p}_j), \\ \hat{p}_j &\in \hat{P}_j, \hat{p}_j \neq \hat{p}_1, \hat{p}_j \neq \hat{p}_2, \end{aligned} \quad (47)$$

where $\hat{p}_j$ is points on E_j which is different from $\hat{p}_1$ and $\hat{p}_2$. The third situation is the containing situation, which is given as: there is no real solutions in (40), however, $\hat{p}_j \in \hat{P}_j$ satisfies

$$[\hat{p}_j, 1]A_l[\hat{p}_j, 1] < 0. \quad (48)$$

or $\hat{p}_l \in \hat{P}_l$ satisfies

$$[\hat{p}_l, 1]A_j[\hat{p}_l, 1] < 0, \quad (49)$$

which means E_j and E_l have containing relationship. The overlap area of E_j and E_l is relatively large corresponding to E_j . Hence, the moving distance d_{jl} is defined as

$$\lambda_{jl} = \max\{a_j, a_l\}, \quad (50)$$

where a_j and a_l are the long axis of E_j and E_l , respectively. The angle θ_{jl} is given as

$$\theta_{jl} = \arccos \frac{x_l^c(a) - x_j^c(a)}{|p_l(a) - p_j(a)|}, \quad (51)$$

where $p_j(a)$ and $p_l(a)$ are the locations of E_j and E_l in $P(a)$ obtained in (13), respectively, and $x_j^c(a)$ and $x_l^c(a)$ are the x-axis coordinates of E_j and E_l .

If there are two real solutions for (42), or (42) exists one real solution (x_1, y_1) , which satisfies

$$[\hat{p}_j, 1]A_l[\hat{p}_j, 1] \geq 0, \quad (52)$$

where $\hat{p}_j \in \hat{P}_j$. The coincidence distance d_{jl} of E_j and $E_l \in E^j(P(e))$ can be obtained by

$$\begin{aligned} [x_1, y_1, 1]A_j[x_1, y_1, 1]^T &= 0, \\ [x_1, y_1, 1]A_l[x_1, y_1, 1]^T &= 0, \\ [x_2, y_2, 1]A_j[x_2, y_2, 1]^T &= 0, \\ [x_2, y_2, 1]A_l[x_2, y_2, 1]^T &= 0, \\ m_{jl} &= \frac{y_1 - y_2}{x_1 - x_2}, \\ y_{m1} &= m_{jl}x_{m1} + b_1, \\ y_{m2} &= m_{jl}x_{m2} + b_2, \\ \lambda_{jl} &= \frac{|b_1 - b_2|}{\sqrt{m_{jl}^2 + 1}} \end{aligned} \quad (53)$$

where (x_1, y_1) and (x_2, y_2) are the points of intersection between E_j and E_l . The slop of line between (x_1, y_1) and (x_2, y_2) is given as m_{jl} . The points of tangents based on m_{jl} can be obtained, while (x_{m1}, y_{m1}) and (x_{m2}, y_{m2}) are two tangent points which have shortest distance. Parameters b_1 and b_2 can be calculated based on m_{jl} , (x_{m1}, y_{m1}) and (x_{m2}, y_{m2}) . The moving distance λ_{jl} can be treated as the overlapped

distance between E_j and E_l , which will be calculated based on m_{jl} , b_1 and b_2 . Angle θ_{jl} is written as

$$\theta_{jl} = \arctan\left(-\frac{1}{m_{jl}}\right). \quad (54)$$

The moving direction of E_j is given as $\vec{u} = [1, 0]$ and $-\vec{u} = [-1, 0]$. Based on $s_1 \times s_2 > 0$, $s_1 \in S_1$, $s_2 \in S_1$, the moving direction of E_j is along x-axis. If all $x_l^c(a) - x_j^c(a) > 0$, all intersections of E_j distribute on the right side of E_j , the projected direction of E_j is $-\vec{u}$. If all $x_l^c(a) - x_j^c(a) < 0$, all intersections of E_j distribute on the left side of E_j , the projected direction of E_j is $\vec{u}$.

If $s_3 \times s_4 > 0$, $s_3 \in S_2$, $s_4 \in S_2$, the intersections distribute above or below E_j , and $|\lambda_j^c|$ can be written as

$$|\lambda_j^c| = \max\{\lambda_{jl}|\sin\theta_{jl}|\}, E_l \in E^j(P(a)), \quad (55)$$

where d_{jl} can be obtained in (50) and (53). The angle θ_{jl} is calculated in (51) and (54). The moving direction of E_j is given as $\vec{v} = [0, 1]$ and $-\vec{v} = [0, -1]$. Based on $s_3 \times s_4 > 0$, $s_3 \in S_2$, $s_4 \in S_2$, the moving direction of E_j is along y-axis. If all $y_l^c(a) - y_j^c(a) > 0$, all intersections of E_j distribute above E_j , and the projected direction of E_j is $-\vec{v}$. If all $y_l^c(a) - y_j^c(a) < 0$, all intersections of E_j distribute below E_j , and the projected direction of E_j is $\vec{v}$. Based on above derivation, the updated position $p_j(e)$ of E_j in $P(e)$ is given as

$$p_j(e) = \begin{cases} p_j(a) - (\mu\lambda_{jl}\cos\theta_{jl} + \varepsilon)\vec{u}, & s_1 \times s_2 > 0 \\ p_j(a) - (\mu\lambda_{jl}\sin\theta_{jl} + \varepsilon)\vec{v}, & s_3 \times s_4 > 0, \end{cases} \quad (56)$$

where E_l is corresponding to (47) Unit vectors $\vec{u}$ and $\vec{v}$ satisfy $\vec{u} = [1, 0]$ and $\vec{v} = [0, 1]$.

Under this situation, $E_l \in E^j(P(e))$ does not change its position based on $P(a)$, which is written as

$$p_l(e) = p_l(a). \quad (57)$$

□

If overlapping areas are scattered around the elliptical agent E_j , the positions for elliptical agent E_j in $P(e)$ can be updated based on the Theorem 5 as follow.

Theorem 5: Let the elliptical agents satisfied (40) be currently positioned in P . Based on $P(a)$ obtained in (13), the positions for elliptical agents E_j and $E_l \in E^j(P(e))$ in $P(e)$ can be obtained as follows:

$$\begin{cases} p_j(e) = p_j(a), \\ p_l(e) = p_l(a) + (\mu\lambda_{jl} + \varepsilon)e_{jl}. \end{cases}$$

where $p_j(e)$ is the position of E_j in $P(e)$ and $p_j(a)$ is the position of E_j in $P(a)$. Vector λ_{jl} is the moving distance of E_j , and θ_{jl} is the projection angle of λ_{jl} to x-axis. The constant parameter μ is used to control the length of moving distance, and $0 < \mu < 0.5$. Coefficient ε is proposed to ensure the moving distance, while e_{jl} is the moving direction for $E_l \in E^j(P(e))$. Unit vectors $\vec{u}$ and $\vec{v}$ satisfy $\vec{u} = [1, 0]$ and $\vec{v} = [0, 1]$.

Proof: Under this condition, for elliptical agent E_j , all intersections of E_j distribute around E_j . In this case,

the position of E_j in $P(e)$ keeps the original position in $P(a)$,

$$p_j(e) = p_j(a), \quad (58)$$

which means $|\lambda_j| = 0$. For $E_l \in E^j(P(e))$, the moving vector λ_l is calculated as

$$\lambda_l = (\mu|\lambda_l| + \varepsilon)e_{jl}, \quad (59)$$

where μ and ε are the positive coefficients, while $|\lambda_l|$ is the length of moving distance for E_l which can be obtained based on (50) and (53). Moving direction e_{jl} of E_l is obtained based on the overlapping situations. If $|\lambda_l|$ is obtained by (50), moving direction e_{jl} is calculated by

$$e_{jl} = \frac{p_l(a) - p_j(a)}{|p_l(a) - p_j(a)|}, \quad (60)$$

where $p_l(a)$ and $p_j(a)$ are the coordinations of center of E_l and E_j in $P(a)$. If $|\lambda_l|$ is obtained by (53), moving direction e_{jl} is calculated by

$$\begin{aligned} e_{jl} &= (\cos\theta_{jl}, \sin\theta_{jl}) \\ &= (\cos(\arctan(-\frac{1}{m_{jl}})), \sin(\arctan(-\frac{1}{m_{jl}}))). \end{aligned} \quad (61)$$

Hence, the updated positions of $E_l \in E^j(P(e))$ are given as

$$\begin{aligned} p_l(e) &= p_l(a) + \lambda_l \\ &= p_l(a) + (\mu\lambda_{jl} + \varepsilon)e_{jl}, \end{aligned} \quad (62)$$

where parameter μ satisfies $0 < \mu < 0.5$. □

According to the proof of algorithm convergence in Section III, the convergence of the two-stage reconfiguration algorithm for elliptical agents can be ensured.

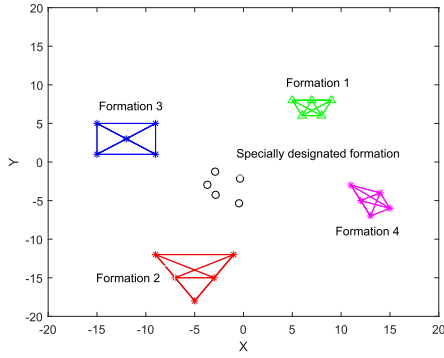
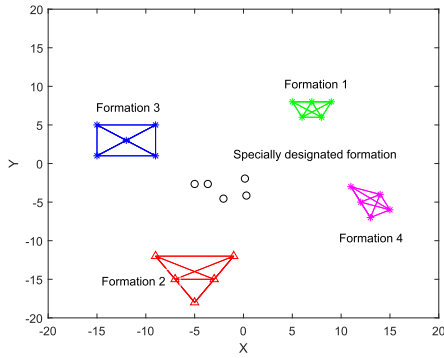
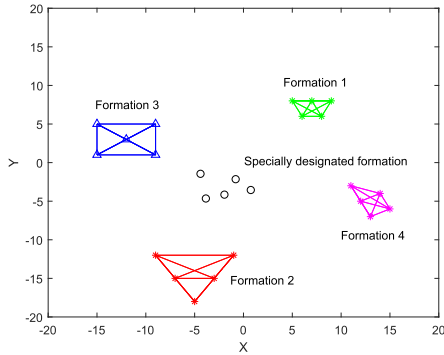
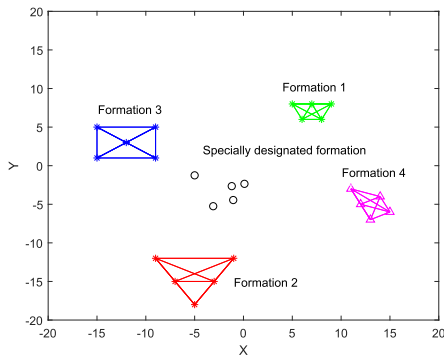
V. SIMULATION RESULTS

In this section, simulation results are given to illustrate the feasibility of the two-stage reconfiguration strategy for a group of agents. Five agents form the multi-agent system, and it is assumed that all agents have the same control strategy. Three examples are provided in the simulation study to demonstrate the effect on constructing the specially designated formation and avoiding overlapping problem with the two-stage reconfiguration strategy. In these examples, all the agents are assumed to have the same dynamics, and undertake the same task in the group.

Example 1: In this example, agents are modeled as dots. Three predefined formations are given as the predefined formations, and make up the predefined formation set F , which are given below

$$\begin{aligned} F^1 &= \{(5 \ 8), (7 \ 8), (9 \ 8), (6 \ 6), (8 \ 6)\}, \\ F^2 &= \{(-9 \ -12), (-7 \ -15), (-5 \ -18), \\ &\quad (-3 \ -15), (-1 \ -12)\}, \\ F^3 &= \{(-15 \ 5), (-12 \ 3), (-15 \ 1), (-9 \ 1), (-9 \ 5)\}, \\ F^4 &= \{(11 \ -3), (12 \ -5), (14 \ -4), \\ &\quad (13 \ -7), (15 \ -6)\}. \end{aligned}$$

The probabilities of the predefined formations F^1, F^2, F^3 and F^4 are given by $q^1 = 0.15$, $q^2 = 0.3$, $q^3 = 0.35$,

Fig. 1. Specially designated formation for five agents located in F^1 .Fig. 2. Specially designated formation for five agents located in F^2 .Fig. 3. Specially designated formation for five agents located in F^3 .Fig. 4. Specially designated formation for five agents located in F^4 .

and $q^4 = 0.2$, respectively, and $q^1 + q^2 + q^3 + q^4 = 1$. The different specially designated formation generated by Algorithm 1 based on different P is given in Figures 1-4.

In these figures, legend $*$ denotes the coordinates of the points in the predefined formations, legend Δ represents the current positions P for these agents, and legend $\circ$ represents the specially designated formation for the system. It can be seen that the specially designated formations have different forms because the current position of each agent leads the different mapping relationships to the possible formations. The corresponding mapping relationships $m_{op}(F)$ to these four predefined formations with $\kappa = 10$ and $\eta = 5$ are given below.

$$m_{op}(F^1) = \begin{Bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 5 & 4 & 3 \\ 5 & 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 & 5 \end{Bmatrix}.$$

$$m_{op}(F^2) = \begin{Bmatrix} 1 & 4 & 2 & 3 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{Bmatrix}.$$

$$m_{op}(F^3) = \begin{Bmatrix} 1 & 3 & 4 & 5 & 2 \\ 1 & 3 & 2 & 5 & 4 \\ 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 5 & 1 \end{Bmatrix}.$$

$$m_{op}(F^4) = \begin{Bmatrix} 1 & 4 & 3 & 2 & 5 \\ 1 & 5 & 4 & 3 & 2 \\ 1 & 2 & 5 & 3 & 4 \\ 1 & 2 & 3 & 4 & 5 \end{Bmatrix}.$$

It can be seen that the different initial positions of five agents will lead the different specially designated formations. This is because of the random mapping algorithm employed in the two-stage reconfiguration strategy.

Example 2: The agents are modeled as circle shapes. Four predefined formations are given to make up the predefined formation set F , which are given below

$$F^1 = \{(20 \ 16), (28 \ 16), (36 \ 16), (24 \ 8), (31 \ 8)\},$$

$$F^2 = \{(-18 \ -32), (-12 \ -40), (-5 \ -48), (-4 \ -40),$$

$$\quad (-1 \ -32)\},$$

$$F^3 = \{(-37 \ 15), (-33 \ 5), (-38 \ -2), (-29 \ -2),$$

$$\quad (-29 \ 15)\},$$

$$F^4 = \{(21 \ -18), (27 \ -23), (30 \ -12), (34 \ -25),$$

$$\quad (23 \ -31)\}.$$

Circular agents in F are not overlapping with the other agents. The radius of each agent is given as

$$r = \{2 \ 3 \ 4 \ 3 \ 2\}.$$

The parameters ε and μ are set as $\varepsilon = 0.2$, $\mu = 0.3$. The specially designated formation for circular agents is illustrated in Figure 5. Circular agents are located in F^1 . The final positions of the whole group are given in $P(c)$, which is shown as

$$P(c) = \{(-14.68 \ -4.9), (-5.29 \ -6.20),$$

$$\quad (4.32 \ -6.20), (-3.20 \ -14.10),$$

$$\quad (-5.61 \ -23.70)\}.$$

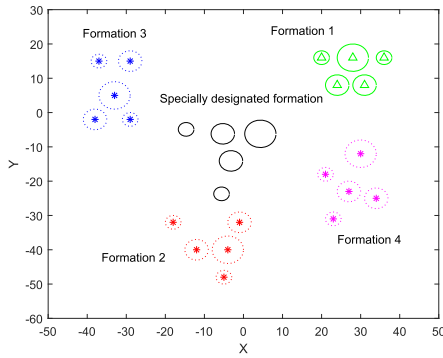


Fig. 5. Specially designated formation for five circular agents located in F^1 .

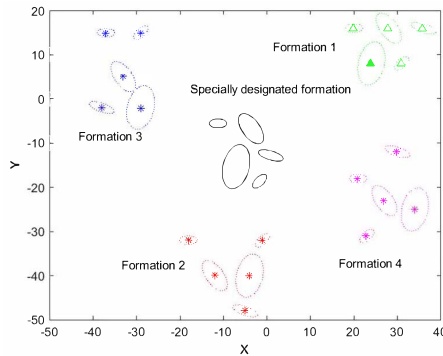


Fig. 6. Specially designated formation for five elliptical agents located in F^1 .

It can be seen that the specially designated formation for these circular agents is constructed without the overlapping problem.

Example 3: In this example, the agents are modeled as ellipses. Four predefined formations are given to form the predefined formation set F , which are same with predefined formation set in Example 2. The elliptical properties of the elliptical agents are

$$\begin{aligned} [a_1, b_1, \theta_1] &= [2, 1, 0^\circ], & [a_2, b_2, \theta_2] &= [4, 2, 127^\circ], \\ [a_3, b_3, \theta_3] &= [3, 1, 341^\circ], & [a_4, b_4, \theta_4] &= [5, 3, 76^\circ], \\ [a_5, b_5, \theta_5] &= [2, 1, 42^\circ]. \end{aligned}$$

Parameters of ε and μ are taken as $\varepsilon = 0.2$, $\mu = 0.3$. The specially designated formation for elliptical agents is illustrated in Figure 6. The agents are located in F^1 . The final position $P(e)$ of each agent is obtained as

$$\begin{aligned} P(e) = \{ & (-11.15 \quad -5.55), (-3.55 \quad -6.75), \\ & (0.99 \quad -12.85), (-7.00 \quad -15.30), \\ & (-1.60 \quad -18.50) \}. \end{aligned}$$

It can be seen that the five agents can find their specially designated formation during idle time by using the two-stage reconfiguration strategy. The specially designated formation is changing based on the mapping relationship between the current positions of these agents and each predefined formation. At the same time, circular agents and elliptical agents can avoid overlapping problems effectively.

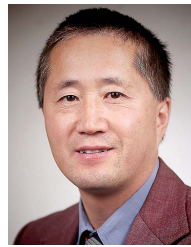
VI. CONCLUSION

In this paper, a two-stage reconfiguration scheme is proposed for a group of agents, which can shorten the expected reconfiguration time when the next command with formation changing is given. The new reconfiguration strategy combined with the random mapping algorithm is developed to find the specially designated formation during idle time based on optimal mappings to predefined formation set. The new scheme is improved for circular agents and elliptical agents to deal with the overlapping problem which may appear in the specially designated formation. Simulations are given to show the effectiveness of the new strategy. In our future work, the varying-probability of each predefined formation will be studied to further improve the flexibility of the established strategy.

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